

Day 1: Basics of C/C++

Topics: Variables, Keywords, Data Types, I/O Functions, Operators

1. Program to Print Hello World

A basic program that outputs "Hello, World!" to the screen.

```
#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    return 0;
}
```

2. Program to Calculate Sum of Two Numbers

Takes two numbers as input from the user and prints their sum.

```
#include <stdio.h>

int main() {
    int a, b, sum;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    sum = a + b;
    printf("Sum = %d\n", sum);
    return 0;
}
```

3. Program to Calculate Area of a Rectangle

Takes length and breadth as input and calculates the area (Area = length x breadth).

```
#include <stdio.h>

int main() {
    float length, breadth, area;
    printf("Enter length: ");
    scanf("%f", &length);
    printf("Enter breadth: ");
    scanf("%f", &breadth);
    area = length * breadth;
    printf("Area of Rectangle = %.2f\n", area);
    return 0;
}
```

4. Program to Calculate Simple Interest

Calculates Simple Interest using the formula: $SI = (P * R * T) / 100$

```
#include <stdio.h>

int main() {
    float principal, rate, time, si;
    printf("Enter Principal: ");
    scanf("%f", &principal);
```

```

    printf("Enter Rate of Interest: ");
    scanf("%f", &rate);
    printf("Enter Time (years): ");
    scanf("%f", &time);
    si = (principal * rate * time) / 100;
    printf("Simple Interest = %.2f\n", si);
    return 0;
}

```

5. Program to Swap Two Numbers

Swaps the values of two variables using a temporary variable.

```

#include <stdio.h>

int main() {
    int a, b, temp;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Before Swap: a = %d, b = %d\n", a, b);
    temp = a;
    a = b;
    b = temp;
    printf("After Swap: a = %d, b = %d\n", a, b);
    return 0;
}

```

6. Program to Convert Temperature: Celsius to Fahrenheit

Converts temperature from Celsius to Fahrenheit using: $F = (C * 9/5) + 32$

```

#include <stdio.h>

int main() {
    float celsius, fahrenheit;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);
    fahrenheit = (celsius * 9.0 / 5.0) + 32;
    printf("Temperature in Fahrenheit = %.2f\n", fahrenheit);
    return 0;
}

```

— End of Day 1 Assignment —